Audio Equipment and Systems

Bachelor's Degree in Audiovisual Systems Engineering

Electroacoustic analogies

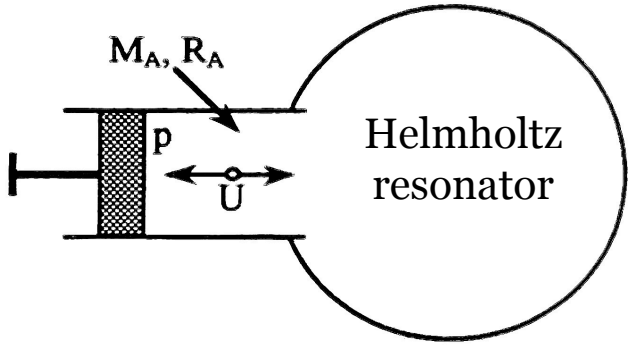
Engineering School
Department of Information
and Communication Technologies

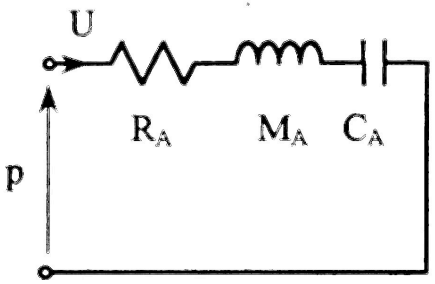
Universitat Pompeu Fabra Barcelona

Adrià García Riber adrian.garcia@upf.edu
Xavier Lizarraga Seijas xavier.lizarraga@upf.edu

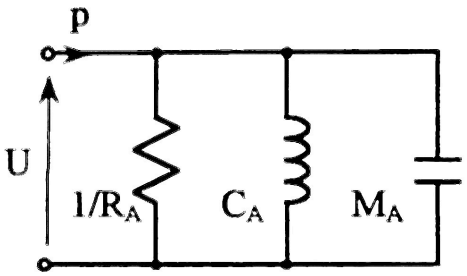
Electro-acoustic analogies

The **compliance** of a gas is its property of changing in volume when subjected to an applied force. It is the reciprocal of stiffness.





Impedance analogy



Mobility analogy

Electric impedance: $V = I \cdot Z \rightarrow Z = \frac{V}{I}$

Acoustic impedance:

$$Z_A = \frac{p}{U} = R_A + j\omega M_A + \frac{1}{j\omega C_A}$$

$[N.s/m^5]$

Electric	Z analogy	Mob. Analogy
Voltage	Pressure (p)	Flow (U)
Current	Flow (U)	Pressure (p)

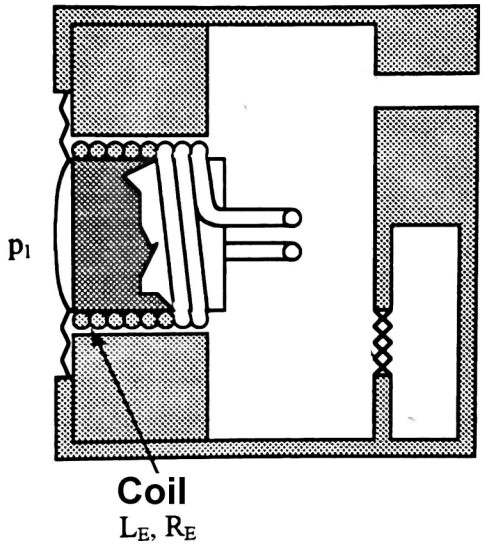
p (Pressure [Pa])

U = S.u $[m^3/s]$ (Volumetric Flow or volumetric velocity)

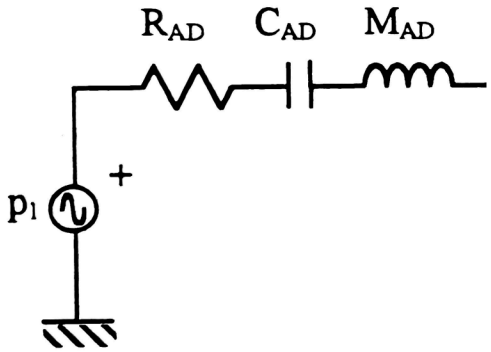
BOTE, José Luis Sánchez. *Transductores electroacústicos*. Fundación General UPM, ETSI Telecomunicación, 2014, p 20.

Electro-acoustic analogy I

Dynamic Microphone



Impedance type circuit



R_{AD} Diaphragm's Acoustic Resistance

C_{AD} Diaphragm's Acoustic Compliance

M_{AD} Diaphragm's Mass

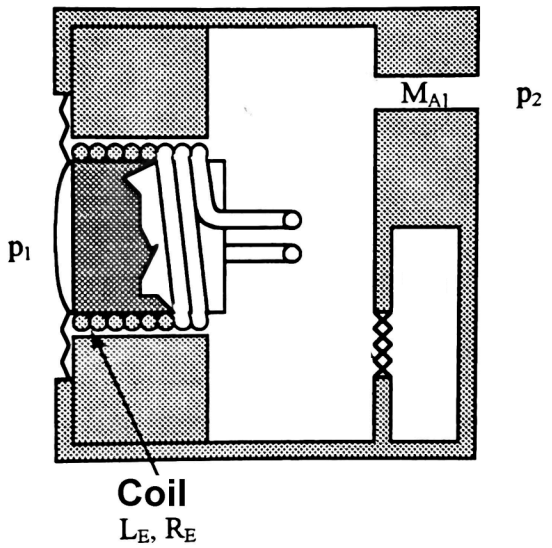
R_E Electric resistance

L_E Electric inductance

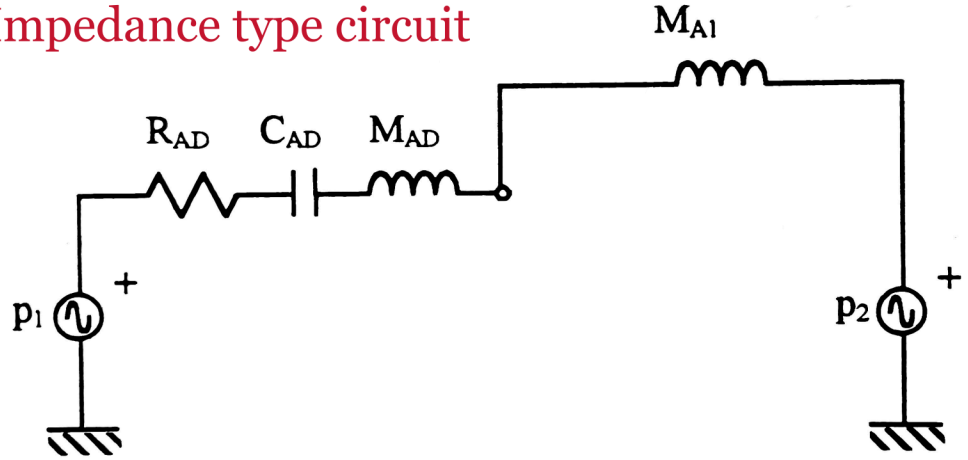
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Electro-acoustic analogy I

Dynamic Microphone



Impedance type circuit



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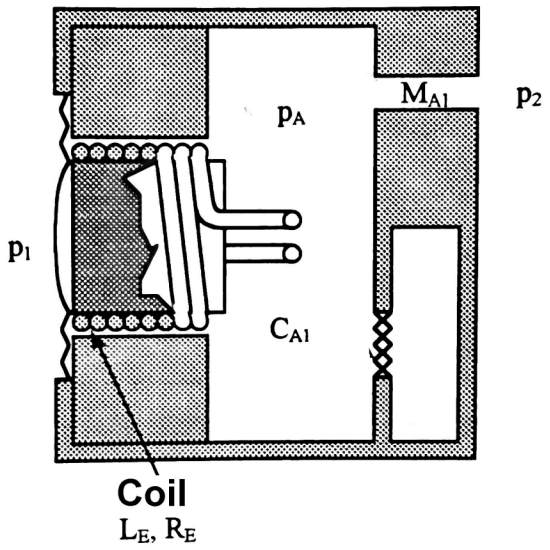
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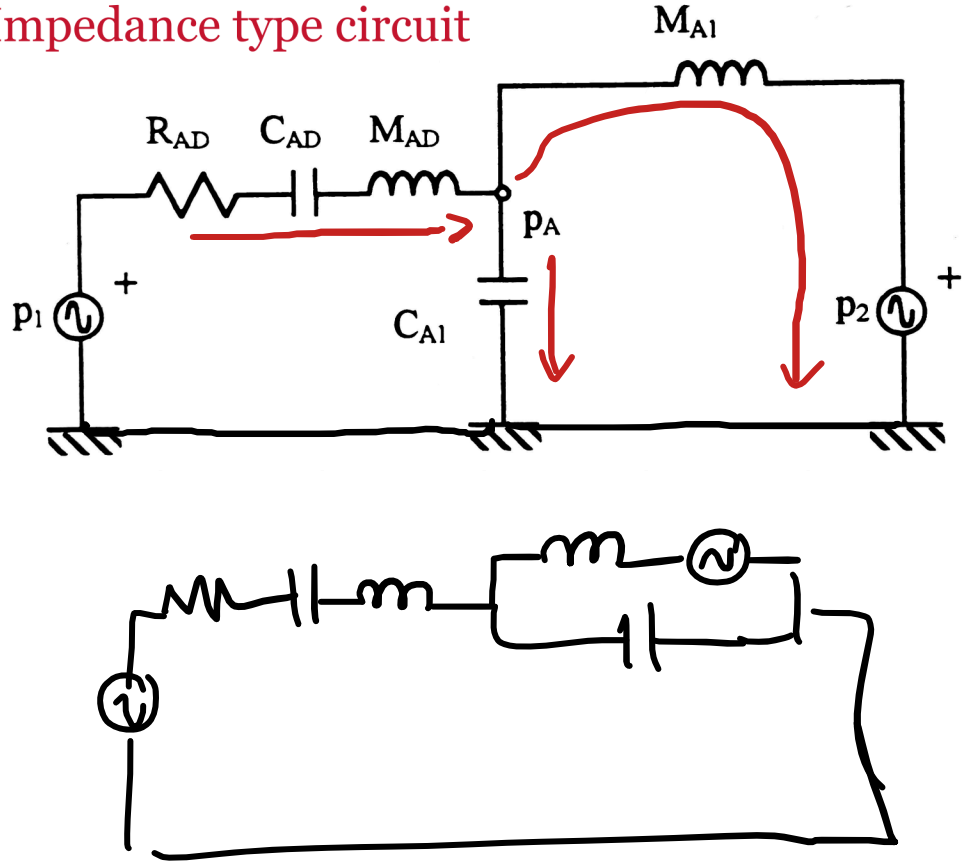
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Electro-acoustic analogy I

Dynamic Microphone



Impedance type circuit



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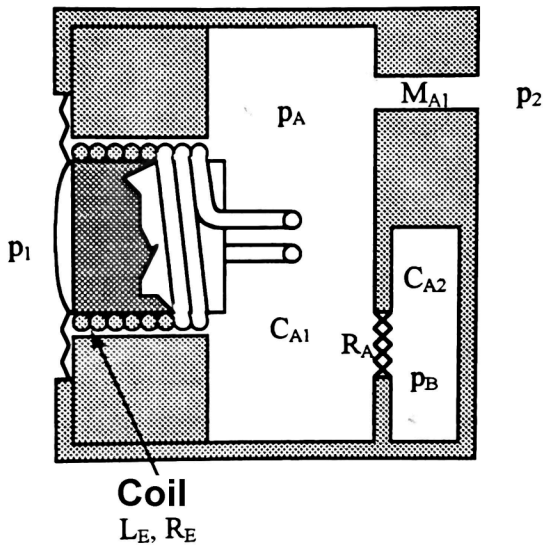
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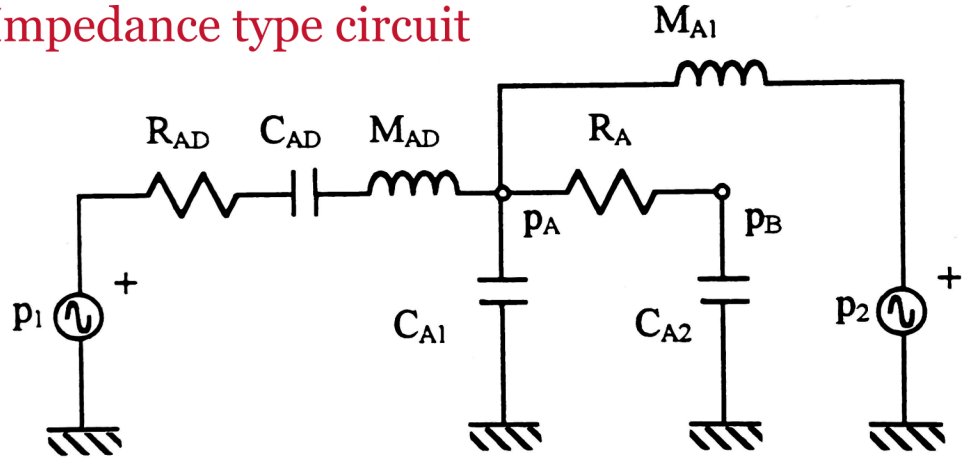
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Electro-acoustic analogy I

Dynamic Microphone



Impedance type circuit



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R_E Electric resistance

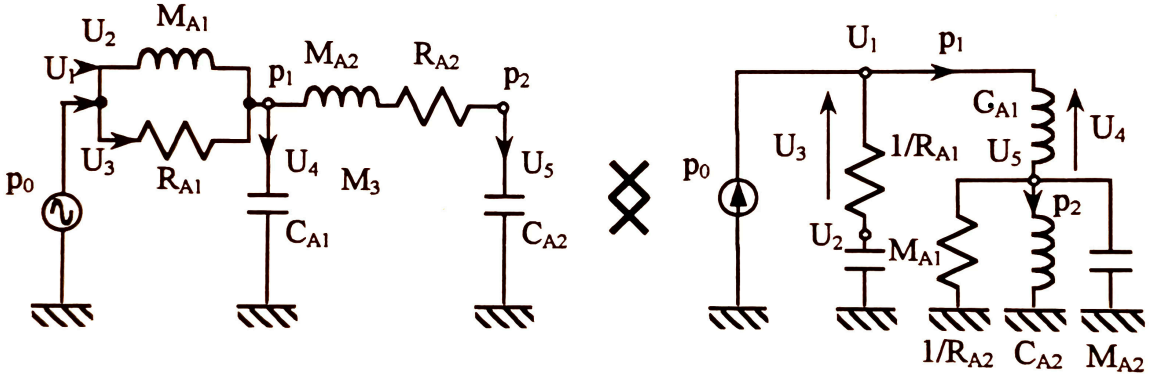
L_E Electric inductance

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Electro-acoustic analogy II

Impedance to Mobility analogy transformations

1. Serial elements transformed into parallel / Parallel into serial
2. Condensers ($C_{M,A}$) transformed into coils ($C_{M,A}$)
   Coils ($M_{M,A}$) into condensers($M_{M,A}$)
   Resistances are transformed into inverse resistances ($R_{M,A}$ <->$1/R_{M,A}$)
3. The sum of currents reaching a knot from elements in parallel are transformed into the sum of voltages of the analogous elements in series, and vice-versa. =Sum of divided currents in parallel turn into divided voltages that get added up in series



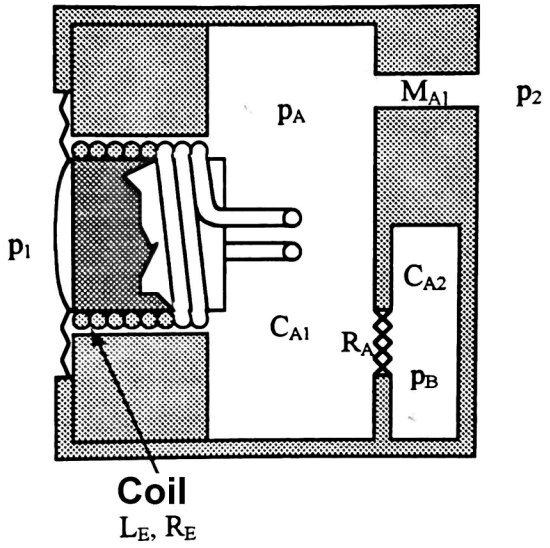
4. Voltage generators are transformed into current generators and vice-versa
5. Transformers invert their transformation rate

1:S to S:1

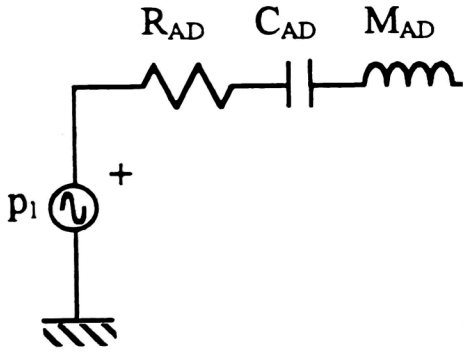
BOTE, José Luis Sánchez. *Transductores electroacústicos*. Fundación General UPM, ETSI Telecomunicación, 2014, p 34.

Electro-acoustic analogy III

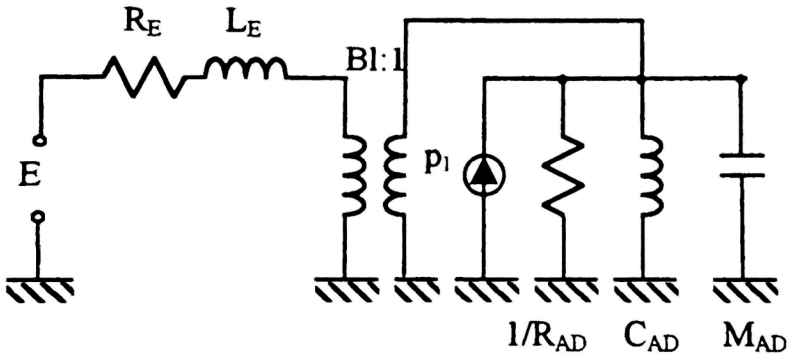
Dynamic Microphone



Impedance type circuit



Mobility type circuit



R_{AD} Diaphragm's Acoustic Resistance

C_{AD} Diaphragm's Acoustic Compliance

M_{AD} Diaphragm's Mass

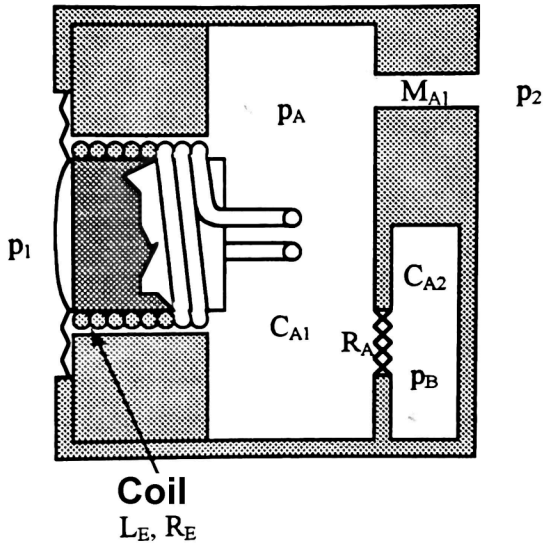
R_E Electric resistance

L_E Electric inductance

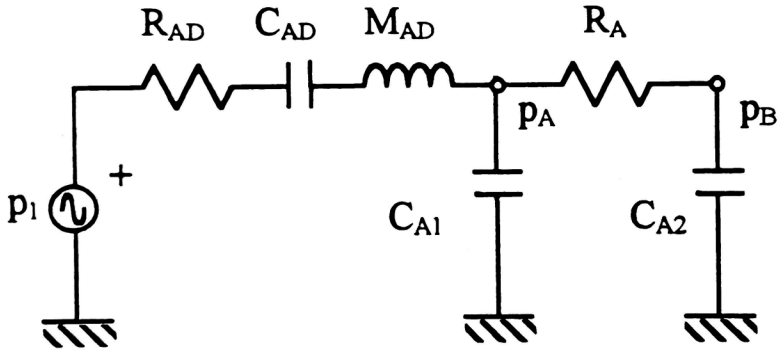
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Electro-acoustic analogy III

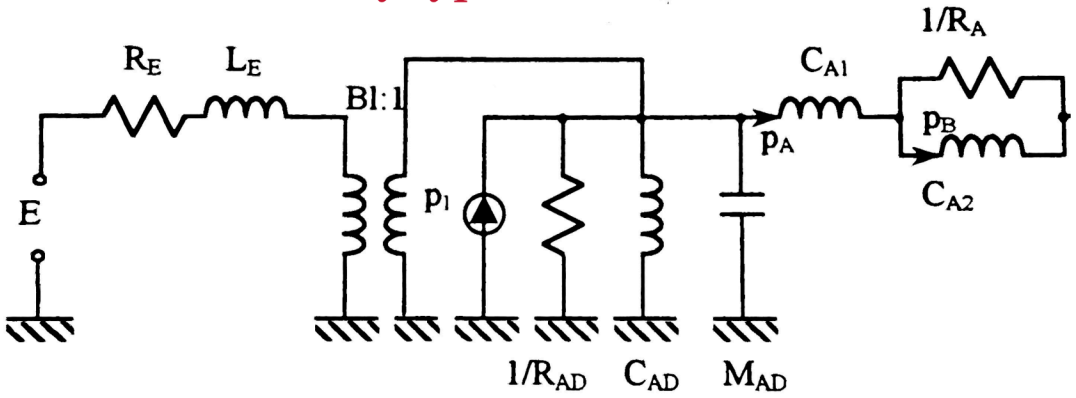
Dynamic Microphone



Impedance type circuit



Mobility type circuit



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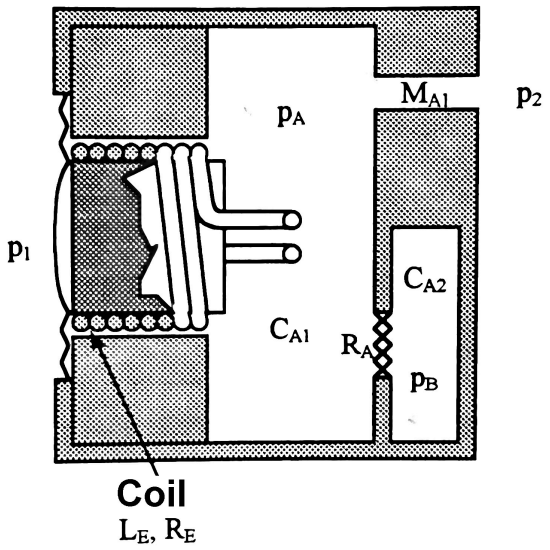
R_E Electric resistance

L_E Electric inductance

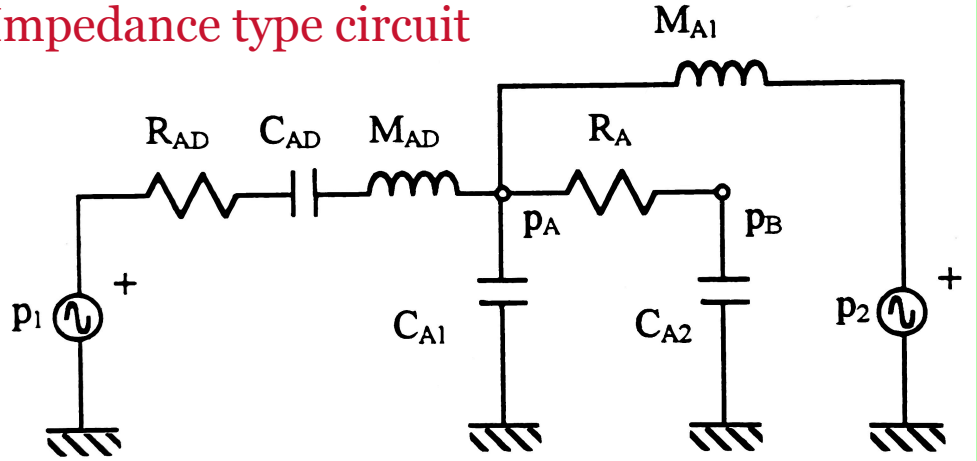
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Electro-acoustic analogy III

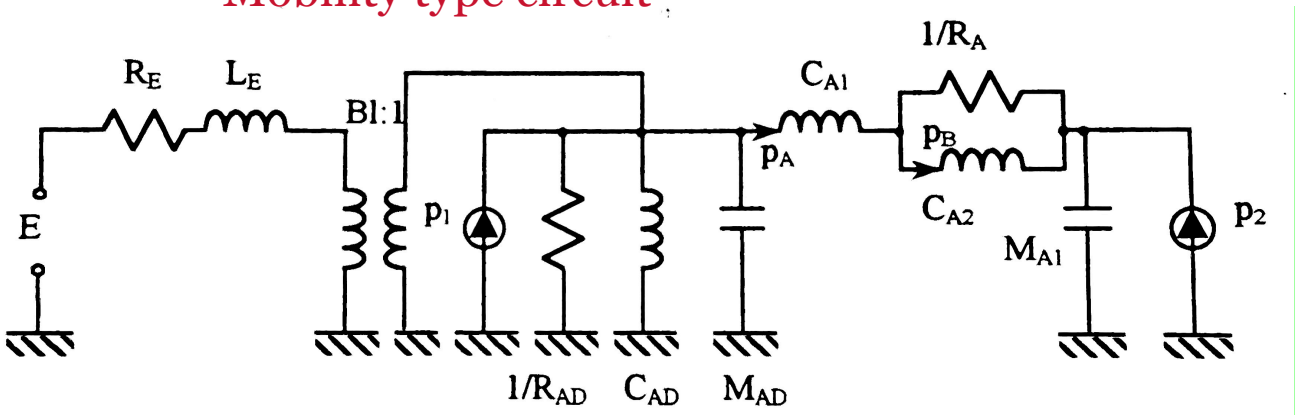
Dynamic Microphone



Impedance type circuit



Mobility type circuit



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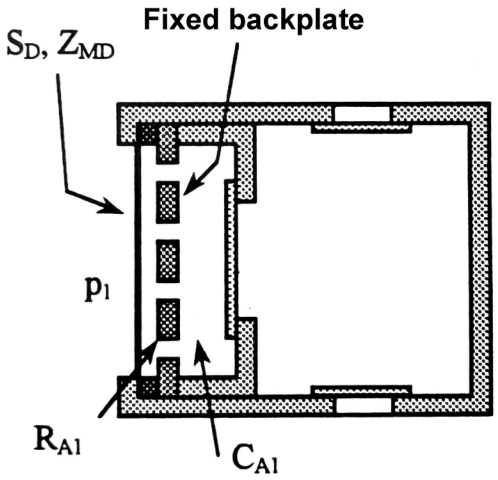
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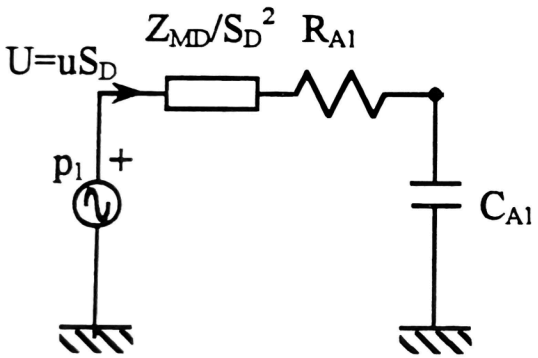
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Electro-acoustic analogy IV

Condenser Microphone



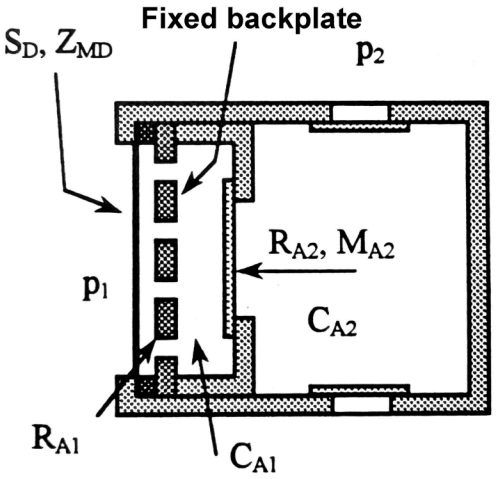
Impedance type circuit



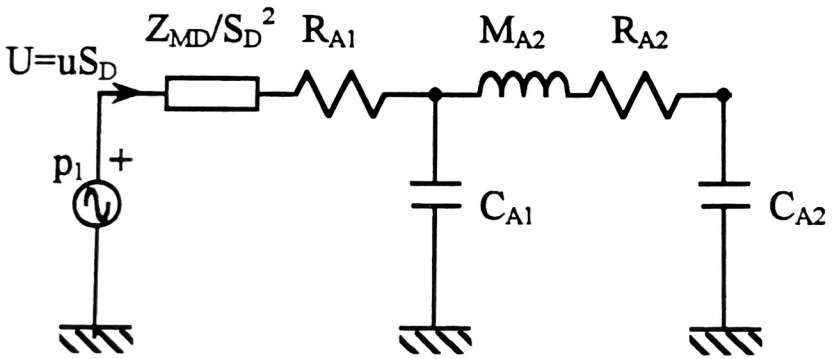
BOTE, José Luis Sánchez. *Transductores electroacústicos*. Fundación General UPM, ETSI Telecomunicación, 2014, p54.

Electro-acoustic analogy IV

Condenser Microphone



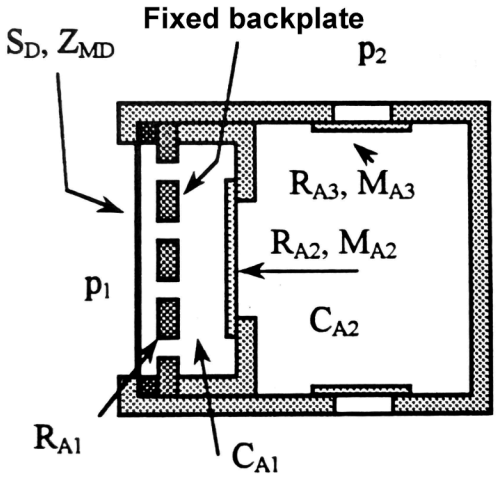
Impedance type circuit



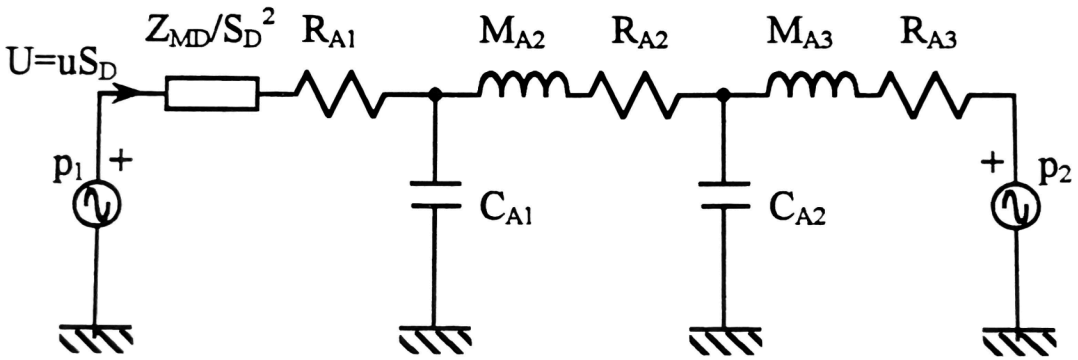
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Electro-acoustic analogy IV

Condenser Microphone



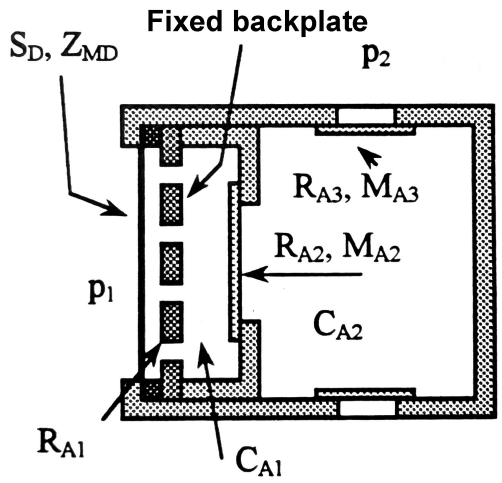
Impedance type circuit



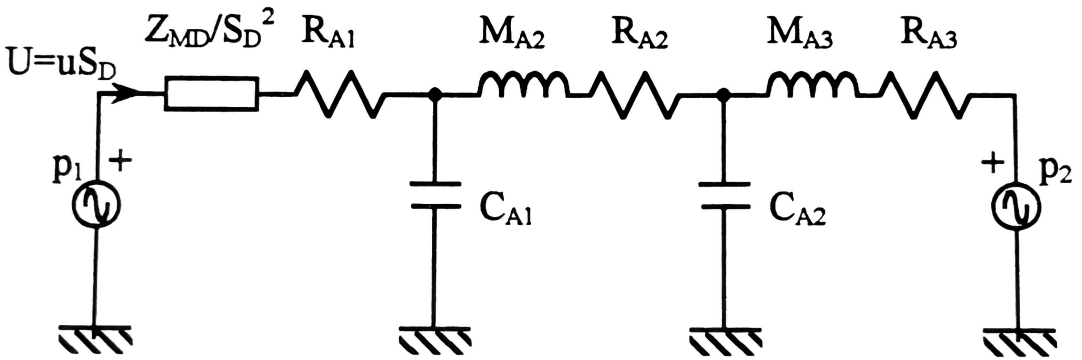
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Electro-acoustic analogy IV

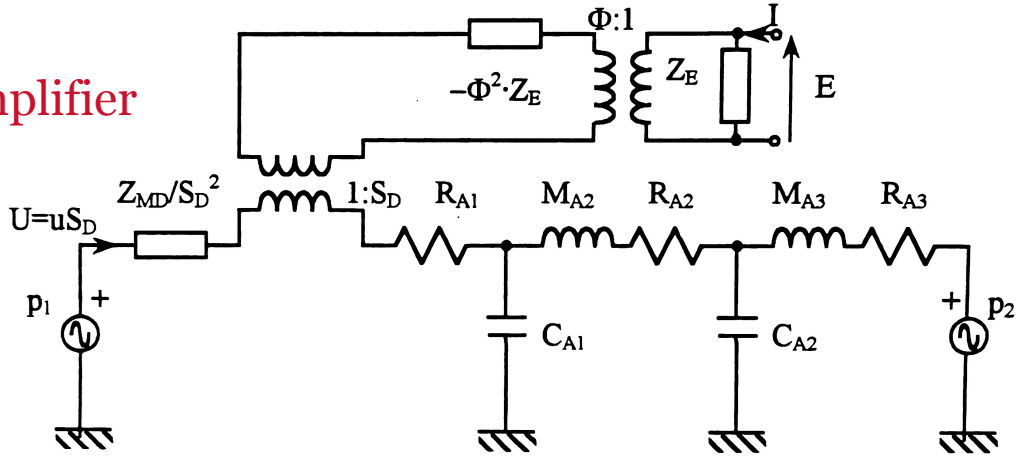
Condenser Microphone



Impedance type circuit



Impedance circuit with preamplifier



E Output voltage

I Intensity

Z_E Electric impedance

$\Phi = B \cdot l$ Magnetic flux [Wb]

B Magnetic flux density [T]

l length of the transformer's coil

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